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FYP REPORT — OFFLINE TASKS CHECKLIST

Things you can do anywhere (phone, tablet, another laptop) — no remote needed

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Author: Chissanupong Saengsint (FYP, U Birmingham)
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INSTRUCTIONS

1. Download the latest report from the remote:
   /home/ubuntu/wish/kuka_kinematic_learner/original/ext_script/scripts/train/
   report_resources/report_drafts/

   FYP_Report_2_Chissanupong_2881058__post_audit_v10_diagrams_20260507_045433.docx

2. Open it in Word (Office, Google Docs, Word Online, or Pages — anything that can edit .docx).

3. Work through the checklist below. Items marked "MANDATORY" must be completed before submission. The rest are polish.

4. When you're done, upload the edited .docx back to the same folder on the remote (or any cloud sync, then I'll fetch it).

5. Tell Claude "I uploaded my edits to <path>" and I'll integrate them with the data-size sweep results (which are still computing on the remote).

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MANDATORY ITEMS

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TASK 1.	Self-assessment box Q1 ("What went well?") Location: near the top of the report, just under the "Project Self-Assessment" heading. It's a 2-cell table; the second cell currently contains the placeholder. Word limit: ~50 words	[10m]

CURRENT PLACEHOLDER TEXT TO REPLACE:

"[Student to complete: ~50 words. Suggested topics to draw on — enjoying the simulation work in Isaac Lab, observing the shared meta-model generalise across DoF configurations, the 99.5% training-time reduction in the 7-DoF case, and the experience of writing the conference paper.]"

WHAT TO WRITE (your own words, ~50):

- What went well?
- One technical thing you found genuinely interesting
- One personal/learning thing you got out of it

DRAFT (you can adapt or write entirely fresh):

"I particularly enjoyed building up the simulation pipeline in Isaac Lab, and I found it satisfying to see a single shared model absorb the kinematic structure of the 5, 6 and 7-DoF iiwa 14 settings.

The 99.5% wall-clock reduction in the 7-DoF case was the most rewarding empirical result. The exercise of preparing a conference paper alongside the FYP also strengthened my technical writing."

(~60 words — feel free to trim or rephrase entirely.)

YOUR FINAL TEXT:

TASK 2.	Self-assessment box Q2 ("What did you find difficult?") [10m] Location: the next 2-cell table, just below Q1. Word limit: ~50 words

CURRENT PLACEHOLDER TEXT TO REPLACE:

"[Student to complete: ~50 words. Suggested topics — depth of learning required for meta-learning literature, interpreting orientation error in quaternion space, scoping the real-robot validation out due to timeline and access constraints, and judgements when the multitask model under-performed the single-task on 7-DoF position.]"

WHAT TO WRITE (your own words, ~50):

- What was difficult? (be honest — not too negative)
- One thing you'd change if you did it again

DRAFT (you can adapt):

"Building the meta-learning intuition required digging through more literature than I expected, and reasoning about orientation error in quaternion space took me time to get used to. I had to scope the real-robot validation out because hardware access and the timeline didn't line up. Looking back, I would have committed earlier to multi-seed evaluation across all three pipeline stages."

(~60 words.)

YOUR FINAL TEXT:

TASK 3.	Appendix D — Generative AI declaration [5m]

Location: Appendix D, Table D.1.
This is MANDATORY by School of Engineering policy.

INTRO PARAGRAPH ABOVE THE TABLE — currently starts with "[Student to confirm and complete this section before submission.]" — DELETE the bracketed phrase so the paragraph begins simply with "The table below lists the generative AI tools used during the project, …".

TABLE D.1 — currently has 3 placeholder rows:

Tool	Used for	Frequency
[Tool name 1]	[e.g., debugging Python errors, …]	[e.g., regularly]
[Tool name 2]	[e.g., explaining concepts during literature review]	[e.g., occasionally]
[Tool name 3]	[e.g., assisting with LaTeX/Word formatting]	[e.g., occasionally]

REPLACE WITH THE TOOLS YOU ACTUALLY USED. Be honest — examiners look at this section. Examples (adapt to your reality):

Claude Code (Anthropic)	Methodology audit of training scripts; proposing data-efficiency / multi-seed experiments; drafting figures and Appendix E listings; rewriting §6.3 to address supervisor feedback. All output reviewed, edited and verified by author.	Heavy in the final week
ChatGPT (GPT-4)	Occasional explanation of meta-learning concepts during literature review; sanity-checking quaternion-distance maths.	Occasionally
GitHub Copilot	Inline code completion in PyTorch scripts.	Regularly
(delete a row if you didn't use that tool — and add rows for any tool I missed)		

YOUR FINAL TABLE — fill in the rows that match what you actually used:

Tool 1: ____________________
Used for: ____________________
Frequency: ____________________

Tool 2: ____________________
Used for: ____________________
Frequency: ____________________

Tool 3: ____________________
Used for: ____________________
Frequency: ____________________

Tool 4: ____________________
Used for: ____________________
Frequency: ____________________

STRONGLY RECOMMENDED

TASK 4.	Read-through Goal: catch any prose that reads oddly, especially in the new sections that were AI-rewritten.	[30m]

Sections to focus on (these are the ones that were significantly rewritten in v8/v9/v10):

- ☐ §3.3 Backbone description (now mentions mask_proj and dropout)
- ☐ §3.5 Eq 3.3 split into 3.3a (Stages 1-2) and 3.3b (Stage 3)
- ☐ §4.4 Per-task standardisation + held-out clarification
- ☐ §4.5 Eq 4.1 — mean Euclidean error (replaces RMSE)
- ☐ §5.4 Multi-seed 4th observation paragraph (with t-test)
- ☐ §6.1 7-DoF interpretation paragraphs (variance acknowledgement)
- ☐ §6.3 Comparison with prior work (3-paragraph rewrite — supervisor's ask)
- ☐ §6.4 Limitations — multi-seed paragraph (precise variance summary)
- ☐ Abstract — multi-seed framing
- ☐ Listings E.1, E.2, E.3 — corrected to match actual code

For anything that reads oddly, just leave a Word comment ("This sentence seems off") — when you upload back, I can iterate on those specifically.

OPTIONAL POLISH

TASK 5.	Cosmetic Unicode pass A few rewritten paragraphs spell "t-hat" / "r-hat" instead of the proper math symbols.	[5m]

In Word's Find & Replace (Ctrl+H):

Find: t-hat   Replace with: t̂   (t with circumflex; can copy from here)
Find: r-hat   Replace with: r̂   (r with circumflex)

If your keyboard can't type the circumflex, leave them as-is — readers will understand "t-hat" perfectly well.

TASK 6.	Replace Figure 5.7 with the 95% CI version (optional)	[1m]

The CI-shaded version of Figure 5.7 is on the remote at:
report_resources/figures/appendix_tb_plots/20260507_045433/fig_multiseed_with_CI_20260507_045433.png

If you sync that folder along with the docx, in Word:

1. Right-click the existing Figure 5.7 image
2. Choose "Change Picture" > "From File"
3. Select the CI-shaded PNG above

The CI-shaded version is more rigorous (shows the 95% CIs we mention in §5.4) but the existing version (just mean ± std bars) is also fine.

TASK 7. Table 6.1 — fill prior-work numbers if you have access [10m]

The cells currently say "(see cited paper; not extracted in this submission)". If you have ScienceDirect / library access, look up:

- KineNN (Diprasetya et al. 2025), Robotics & Computer-Integrated Manufacturing, vol. 94, p. 102945. Find their reported position / orientation error on UR5 forward kinematics. Replace cell.

- Cursi et al. 2022 — search "Augmented NN in SE(3)" on Google Scholar. Need to add a reference number to bibliography too if you fill this row.

- Köker et al. 2004, Robotics & Autonomous Systems vol. 49(3-4), pp. 227-234. They report IK joint errors (not FK end-effector). If you can map IK joint error to end-effector error via DH parameters, fill the row; otherwise consider dropping the row.

If you can't find the numbers in 10 minutes, leave the cells as-is — the table caption already explains why they're not transcribed.

ALL DONE?

When you've finished all the items you want to do:

1. Save the docx (in Word: File > Save).

2. Upload it back to the remote. Either:
   a) scp / rsync to /home/ubuntu/wish/kuka_kinematic_learner/original/ ext_script/scripts/train/report_resources/report_drafts/ FYP_Report_2_Chissanupong_2881058__YOUR_EDIT.docx
   b) Drop on a cloud drive that the remote can read

3. Tell Claude (in any future session): "I uploaded my edits to <path>"
   Claude will:
   - Diff your edits against v10
   - Apply data-size sweep results (Fig 5.8) on top of your edits
   - Save as v11_dataeff_with_user_edits

WHAT IS HAPPENING ON THE REMOTE WHILE YOU'RE AWAY:

- The data-size sweep is running in the background (~3.7 hours)
- When it finishes (~15:00), Claude will automatically:
  - Render Figure 5.8 (data-efficiency curve)
  - Apply it to whichever docx is the latest
  - Report the 7-DoF verdict
- If the machine reboots, see CONTINUATION_PLAN.md for recovery

DO NOT CLOSE THE LAPTOP IF YOUR LAPTOP IS THE GPU MACHINE — the sweep needs the GPU running. If you must, configure the lid action to "do nothing" or "sleep display only" first.

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END OF FILE
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